

Business Requirements Document (BRD)

Project Title: Murex Integration with MarkitWire Trade Affirmation & STP Platform

Document Reference: BRD-MX-MW-STP-001

1. Document Overview & Context

1.1 Purpose

This document specifies the business and technical requirements for integrating **Murex** with the **MarkitWire** trade affirmation platform. The integration enables automated Straight-Through Processing (STP), pre-trade affirmation, Central Counterparty (CCP) clearing routing (LCH/CME), rate transition handling, and message replay capabilities for non-connected test environments.

1.2 Target Audience

- Business Analysts & System Integrators
- Murex Technical Consultants
- Middle and Back Office Operations
- Quality Assurance & Testing Teams

2. System Comparison & Functional Roles

2.1 System Architecture Roles

- **MarkitWire (Front-End Affirmation):** Web-based trade affirmation platform. Operates as a layer above Murex to reduce operational risk before trade details hit downstream risk systems. Both counterparties must log into MarkitWire to review, accept, or affirm trade details.
- **Murex (Downstream Trade Capture & Processing):** Operates as the internal trade capture, risk, and back-office management system. Does not support direct external counterparty logins. Once a deal is affirmed and booked, Murex generates post-trade confirmation outputs (e.g., SWIFT or HTML).

2.2 Functional Matrix

Feature / Dimension	MarkitWire Screen / Interface	Murex Screen / Interface
Primary System Layer	Web-based Front-End Affirmation & Pre-Risk Capture	Downstream Trade Capture, Risk, & Back-Office Engine
User Access Model	External/Dual-Party access	Internal Access Only (Issued to

Feature / Dimension	MarkitWire Screen / Interface	Murex Screen / Interface
	(Traders from both counterparties log in)	internal bank traders and back-office ops)
Core Screen Features	Ticket booking wizards, counterparty search, product templates, Clearing auto-send checkboxes	Smart Qs, Contract Data Viewers, Exchange Task Monitoring, Error Queues
Output Data Representation	Plain-English financial deal terms, SWML XML tree view	MXML Contract Structure, Contract ID generation, UDF Labels
Confirmation Role	Pre-trade affirmation prior to risk engine delivery	Post-booking output generation (SWIFT / HTML)

3. Screen Layouts & User Interfaces

3.1 MarkitWire User Interface (MW UI)

The MarkitWire web screen is used for manual and automated trade entry, counterparty selection, deal term customization, and clearing assignment.

Key MW Screen Elements:

1. **Trader & Counterparty Selection:** Drop-down lists and lookup modals for picking trading entities (e.g., Mega Bank) and trader profiles.
2. **Deal Structure Controls:** Inputs for product type (e.g., Interest Rate Swap), currency, tenor, and fixed rate.
3. **Clearing House Integration Panel:** Options to select clearing venues (e.g., LCH Clearnet) and toggle the Auto send for clearing checkbox.
4. **Action Toolbar:** Controls to perform/submit trades, duplicate deals, withdraw unaccepted deals, or view underlying SWML data.
5. **Internal Data Tab:** Displays mapped upstream broker IDs (e.g., TradeWeb) and receives the Murex Contract ID back via integration.

3.2 Murex User Interface (MX UI)

Key MX Screen Elements:

1. **Exchange Task Sheets:**
 - **MW Inbound:** Captures SWML payloads and converts them into MXML for trade insertion and events.
 - **MW Outbound:** Sends the Murex Contract ID back to MarkitWire's Internal Data tab.
 - **ITF Error Management:** Tracks generic integration errors.
2. **Smart Queues Monitor (Market Wire Error Q Monitor):** Centralized view listing processed messages, active errors, workflow history, and underlying deal payloads.

3. **Formula & Mapping Tables:** Configuration interfaces mapping SWML variables to Murex User-Defined Fields (UDFs), deal generators, and user IDs.

4. Trade Origin Identification Logic (ID Suffixes)

To maintain execution auditability across screens, Murex appends numerical suffixes to the standard MarkitWire Trade ID within reference fields:

- 1 **Suffix (Internal Trader-Initiated):** Assigned to deals directly booked by an internal trader against an external counterparty.
- -2 **Suffix (External / Clearing House-Initiated):** Assigned to deals initiated by external counterparties or clearing houses (e.g., LCH transition deals).
- -3 **Suffix (Broker Platform & Electronic Executions):** Assigned to deals originating from upstream electronic venues/brokers (e.g., Bloomberg, TradeWeb), Swap Execution Facilities (SEF), Multilateral Trading Facilities (MTF), and drop copy notifications.

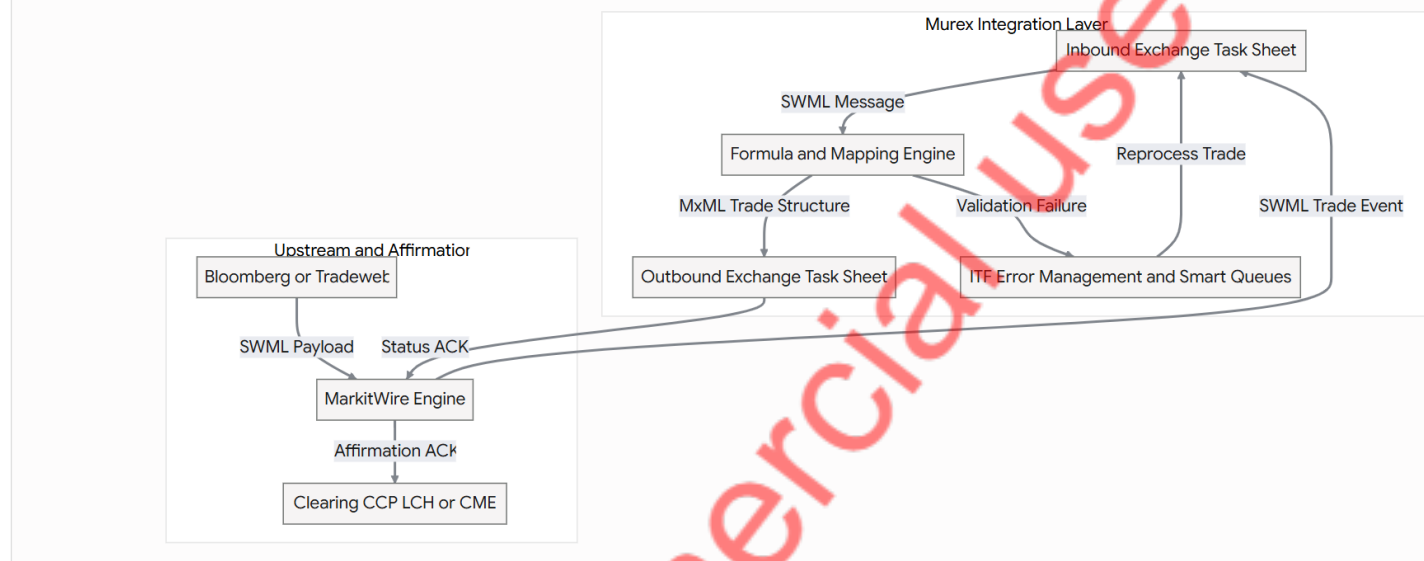
5. Scope of Transaction & Deal Types

The integration processes the following transaction types:

1. **Bilateral Deals:** Direct non-cleared trades between two institutions.
2. **Clearing Deals:** Trades assigned automatically or manually to Central Counterparties (CCPs) like LCH or CME.
3. **Broker Deals:** Trades originating from upstream broker platforms (e.g., TradeWeb, Bloomberg).
4. **Drop Copy Messages:** System notification messages copying pre-cleared trade details from trading venues without re-clearing.
5. **Single-Sided Deals (SSP):** Transactions where the counterparty lacks a MarkitWire account; MarkitWire auto-accepts the deal, and Murex issues post-trade confirmations.
6. **Inter-Entity Deals:** Internal transactions booked between different branches or entities of the same institution.

Detailed Data Flow & Architecture

Murex & MarkitWire Trade Architecture



7. Technical Configuration, Recovery, & Offline Testing

7.1 Outage & Historical Recovery Processing

- **Same-Day Listener Reconnect:** Handled automatically by the persistent listener connection (Sync Subscribe For All), which retrieves missed same-day messages once restored.
- **Multi-Day Database / System Outages:** Requires running explicit historical **Sync Queries** across date/time ranges to retrieve missing transition and cash compensation payloads.

7.2 Clearing House (LCH) Rate Transitions

- **Unwind Processing:** Pre-existing deals must be unwound prior to inserting replacement deals.
- **Inactive Trader User Mapping:** Inactive or departed traders with active deals must be dynamically mapped to active system/treasury user profiles to prevent unwind processing failures.
- **Cash Compensation Deals:** Single cash flow (SCF) balance trades sent by LCH to offset transition P&L variances must be automatically routed to designated treasury portfolios and allocated to active application service users.

7.3 Shared Environments & Offline Message Replaying

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- **Central Export Directory (MW export message):** Connected test environments copy incoming SWML payloads to a shared network directory, allowing multiple test instances to share real-time feeds without separate physical MarkitWire connections.
- **Local Message Replaying:** Non-connected environments use local file-reader input directories to process copied SWML files.
- **Recovery Mode Configuration Requirement:** Prior to replaying local files, the configuration flag **UDF6 must be explicitly set to FALSE**. This bypasses recovery mode state logic, ensuring replayed messages enter Murex as new trade insertions rather than failing state checks.

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